

Air Supply System Model with Exhaust Gas Recirculation for Improving the Life of Fuel Cell

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Abstract—This paper presents an air supply system model with the exhaust gas recirculation (EGR) function. High potential is one of the key factors for degradation of proton exchange membrane fuel cell. The harmful high voltage always appears when the fuel cell works at small load. The output voltage is greatly affected by the oxygen stoichiometric at the cathode. The high voltage can be lowered down by controlling the oxygen stoichiometric in a suitable range. Based on this concept, an air supply system model with EGR is proposed and modeled. A feasible control strategy for lowering the output voltage is also deployed. Simulation results of the model show that, the oxygen ration in the inlet air can be greatly reduced, as well as the output voltage.

Keywords—proton exchange membrane fuel cell; exhaust gas recirculation; high voltage; oxygen stoichiometric; control strategy

I. INTRODUCTION

Nowadays, due to the fast development of economy, we are facing great energy and environment problems.

The past and future trends for the consumption and discovery of conventional oil had been conducted. And the results showed that consumption exceeded discovery in 1980 and it is unlikely that there will be any sudden increase in the future.^[1]

According to [1], the increase of CO₂ emission will cause lots of environment problems, such as the rise of the sea level and the rise of the global temperature. So, the necessity for immediate cuts in CO₂ is an urgent and long-term issue.

Therefore, we desperately need a revolution in vehicle energy source. Proton exchange membrane fuel cell (PEMFC) has been considered an ideal alternative for internal combustion engine in recent years because of its typical advantages like high efficiency, zero emission and low temperature operation. However, PEMFC durability, especially under a wide range of operational conditions, is still one of the current limitations preventing its wide scale commercialization as a practical power source.^[2] Thus great efforts have been made to improve the durability of PEMFC.

In [3], lots of degradation mechanisms and mitigation strategies are summarized. The failure modes based on electrochemistry of different components, like membrane, catalyst layer, gas diffusion layer, are described in detail and the mitigation strategies are promoted based on material improvement.

The effect of humidity and oxygen partial pressure on degradation of catalyst is studied in [4]. The result informs us

that the humidity control during operation is quite important to extend the life of PEMFC.

Research conducted by the Japanese, mainly by Nissan and Toyota, show that the high voltage is a mortal threat to the durability of PEMFC. During load cycling and idling operation, changes in the electric potential and the holding of a high potential oxidize and reduce Pt. As a result, the surface of some Pt particles dissolves to produce an ion state. The dissolution of Pt reduces the Pt surface area compared with the initial condition, which causes performance to decline.^[5] And test data presents in [1] tells that the higher the voltage is, the shorter the PEMFC life is. When the voltage is ranging from 0.65V to 0.85V, the life of PEMFC is six times longer than the case of ranging from 0.65V to open circuit voltage.

A typical polarization curve is showed in figure 1. The data comes from the test of the fuel cell from Sunrise in 2012, the rated power of this stack is 50 kW. We can clearly see from the curve that when the fuel cell is operating at low current density, the potential is high, especially when the current density is lower than 100 mA/cm². According to [6], the reason of this phenomenon is that the activation loss, ohmic loss, concentration loss are all decreased with the decrease of current density. Especially, the activation loss is quite nonlinear, it will decrease sharply with the current density at low load condition.

The stack is made of 520 cells, and each cell has a 300cm² membrane. So the output power of the stack is more than 12 kW, which is about 25% of the rated power of the stack, when the current density is 100 mA/cm². Figure 2 shows the time weighting of different load rate for a fuel cell engine used for the bus which is worked for the World Expo in Shanghai.

The data in figure 2 shows that there is a big chance that the load rate of the fuel cell is under 20%. So the harmful high potential condition lasts long during working. Thus, the optimization is necessary.

The study in this paper focused on getting the output voltage under control. First, the mechanism of high voltage and the mitigation strategy are introduced and the concept of using oxygen stoichiometric to get the output voltage under control is promoted. Then an air supply system model with EGR which can flexibly control the amount of air and oxygen is built. Finally, a feasible control strategy based on the model is designed.

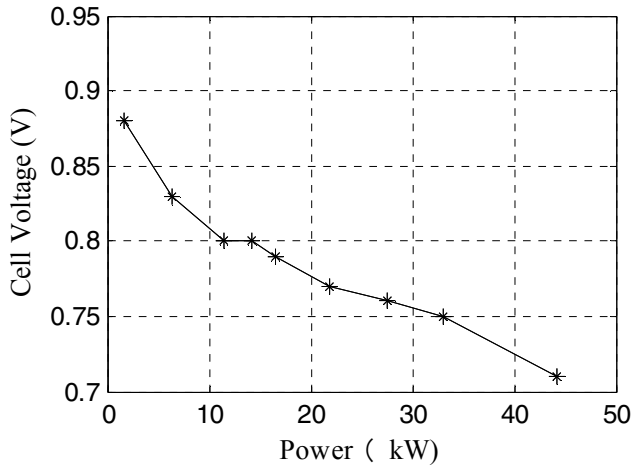


Fig1. A typical polarization curve of PEMFC

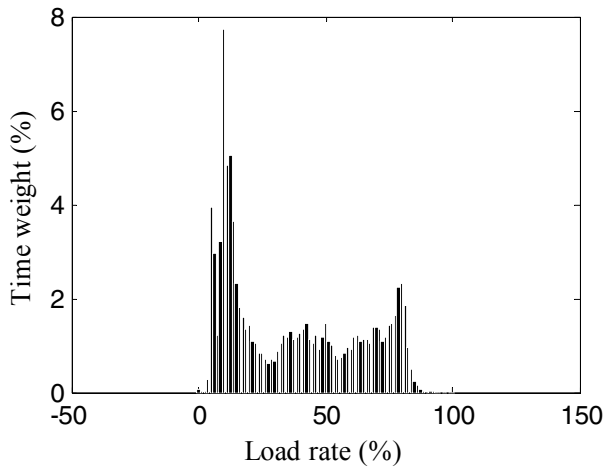


Fig2. Time weighting of different load rate

II. MODELING PROCEDURE

[7] tells the relationship between air stoichiometry and concentration loss. When the stoichiometry of air is smaller than 1.1, the concentration loss rises quickly.

However, the ability of taking away the generated water dropped with the reducing of air supply, so flooding at cathode flow channel may occur. Therefore, in order to achieve the goals of both lowering the potential and avoiding the flooding, the amount of oxygen and air supply should be decoupled. The decoupling could be realized by adding exhaust gas recirculation to the air supply subsystem, the simplified physical model is showed in figure 3.

By using EGR, the amount of oxygen could be changed without changing the amount of air. Also, the amount of oxygen could be flexibly controlled by accurately control the opening area of the two throttles.

This model is focused on the air supply subsystem, so the characteristics of other subsystems like hydrogen supply system are ignored. To simplify the mathematic model and

emphasize the key point, some reasonable assumptions are listed below:

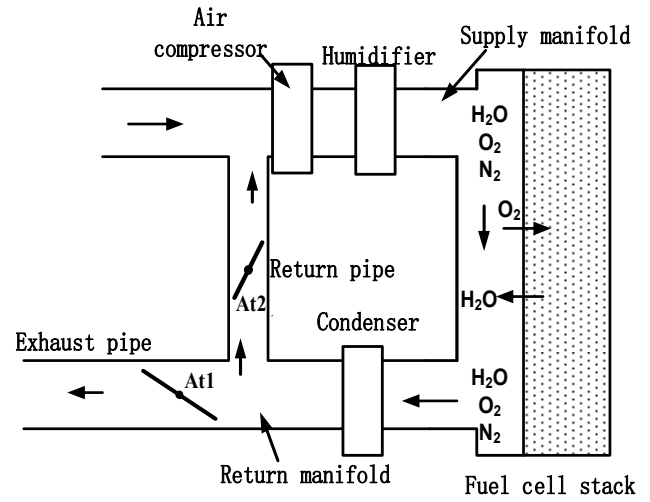


Fig3. Simplified air supply model with EGR

1. The ideal gas law is employed for the gas mixture.
2. The effect of the cooling subsystem is ignored, so the temperatures of the whole system are regarded as a constant.
3. The effect of hydrogen supply subsystem is ignored, the pressure of anode channel is considered as the same with the cathode channel.
4. The humidifier and condenser are regarded as ideal components, so the humidity can accurately satisfy the demand.
5. Lumped model is used for every cavity.

Nomenclature	
R	universal gas constant
P	pressure(bar)
t	time(s)
V	volume(m ³)
W	mass flow rate(g s ⁻¹)
T	temperature(K)
M	gas molar mass(kg mol ⁻¹)
At	throttle opening area(m ²)
k	ratio of specific heat of air
I	current(A)
F	faraday constant(96485C mol ⁻¹)
m	mass(kg)
x	molar fraction
λ	stoichiometric ratio
Subcripts	
in	inlet
out	outlet
ca	cathode
an	anode
ds	down stream
us	upstream

air	air
atm	atmosphere
O ₂	oxygen
H ₂	hydrogen
H ₂ O	water
vap	vapor
sat	saturation
sm	supply manifold
rp	return pipe
cp	compressor
hum	humidifier
con	condenser

The pressure equation of every cavity can be expressed as^[8]

$$\frac{dP}{dt} = \frac{RT}{VM} (W_{in} - W_{out}) \quad (1)$$

The throttle flow equation can be expressed as^[8]

$$W = \frac{C_D A t P_{us}}{\sqrt{RT}} \left(\frac{P_{ds}}{P_{us}} \right)^{\frac{1}{k}} \left\{ \frac{2k}{k-1} \left[1 - \left(\frac{P_{ds}}{P_{us}} \right)^{\frac{k-1}{k}} \right] \right\}^{\frac{1}{2}} \quad (2-1)$$

when $\frac{P_{ds}}{P_{us}} > \left(\frac{2}{k+1} \right)^{\frac{k}{k-1}}$, or

$$W = \frac{C_D A t P_{us}}{\sqrt{RT}} \times k^{\frac{1}{2}} \left(\frac{2}{k+1} \right)^{\frac{k+1}{2(k-1)}} \quad (2-2)$$

when $\frac{P_{ds}}{P_{us}} \leq \left(\frac{2}{k+1} \right)^{\frac{k}{k-1}}$.

A. Air compressor model

The air compressor model from [9] is used in this model. Neural network technique is used for MAP interpolating and extrapolating. However, the model represented in [9] used the downstream pressure and the motor voltage as inputs, so the air flow rate could not automatically satisfy the demand. The relationship between the amount of reacted oxygen and the output current can be expressed as^[8]

$$m_{O_2, reacted} = M_{O_2} \times \frac{n \times I}{4F} \quad (3)$$

where n stands for the number of cells. Thus, the amount of air needed can be calculated with the expression below:

$$m_{air} = \frac{\lambda \times m_{O_2, reacted} \times M_{air}}{M_{O_2} \times 0.21} = \frac{\lambda \times M_{air}}{0.21} \times \frac{n \times I}{4F} \quad (4)$$

Based on (3) and (4), a PI controller was added to the compressor model to change the input from motor voltage to the output current. Therefore the air amount can be accurately controlled. And by setting the value of λ , the model can satisfy different working condition. The control effect is

showed in figure 4. The red line stands for the demanded air amount and the green line stands for the actual air amount.

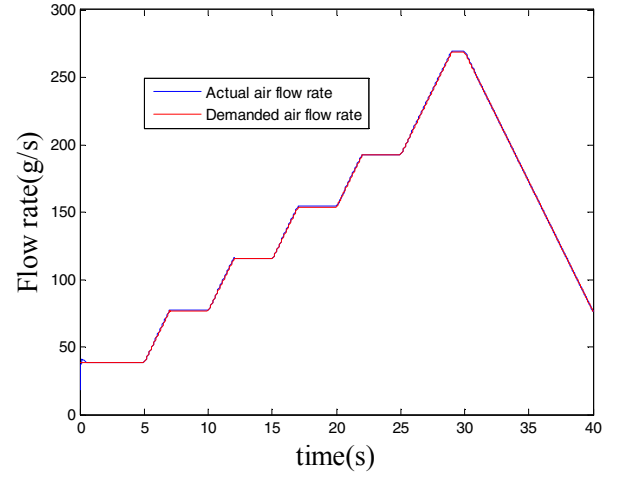


Fig.4 The control effect of the PI controller for the compressor

B. Gas mixture model

Because of the import of EGR, the molar mass of the gas being sucked into the compressor M_{mixair} is different from M_{air} . The calculation of M_{mixair} will be described later.

The molar fraction of vapor in the gas after the compressor and before humidifier can be expressed as

$$x_{vap_atm} = \frac{P_{sat}(T_{atm}) \times RH0}{P_{atm}} \quad (5)$$

where $P_{sat}(T_{atm})$ means the saturated vapor pressure at ambient temperature, $RH0$ means the humidity of the atmosphere. The molar fraction of vapor in the gas after the humidifier can be expressed as

$$x_{vap_hum} = \frac{P_{sat}(T_{hum}) \times RHc}{P_{sm}} \quad (6)$$

where $P_{sat}(T_{hum})$ means the saturated vapor pressure at the set temperature, RHc means the set humidity.

Usually, compared to x_{vap_hum} , x_{vap_atm} is quite small, so the total flow rate would not be influenced much if we just put x_{vap_atm} to zero. Thus, the followed modeling process would ignore the effect of x_{vap_atm} .

The flow rate after the humidifier can be expressed as

$$W_{hum_out} = W_{hum_in} \times \frac{(1-x_{vap_hum}) \times M_{mixair} + x_{vap_hum} \times M_{h_2o}}{(1-x_{vap_hum}) \times M_{mixair}} \quad (7)$$

where W_{hum_in} is flow rate that comes from the compressor.

Then, after the humidifier, there is no gas mixture in the supply manifold, but in the stack, the component of gas should be paid grate attention to.

Due to the effect of EGR, the flow rate of inlet oxygen is connected to the flow rate and oxygen molar fraction of return pipe gas. So the expression can be written as

$$W_{o_2_ca_in} = W_{hum_in} \times \left(\frac{W_{rp}}{W_{cp}} \times \frac{x_{o_2_rp} \times M_{O_2}}{x_{o_2_rp} \times M_{O_2} + (1 - x_{o_2_rp}) \times M_{N_2}} + \frac{1 - W_{rp}}{W_{cp}} \times \frac{0.21 \times M_{O_2}}{M_{air}} \right) \quad (8)$$

Similarly, the flow rate of nitrogen can be expressed as

$$W_{N_2_ca_in} = W_{hum_in} \times \left(\frac{W_{rp}}{W_{cp}} \times \frac{(1 - x_{o_2_rp}) \times M_{N_2}}{x_{o_2_rp} \times M_{O_2} + (1 - x_{o_2_rp}) \times M_{N_2}} + \frac{1 - W_{rp}}{W_{cp}} \times \frac{0.79 \times M_{N_2}}{M_{air}} \right) \quad (9)$$

Here, the humidity of the gas in the stack is assumed to be saturated. Thus the total mass of gas in the cathode channel can be expressed as

$$m_{ca} = \int (W_{o_2_ca_in} - W_{o_2_ca_out} - W_{o_2_con}) + \int (W_{N_2_ca_in} - W_{N_2_ca_out}) + \frac{P_{ca_sat} \times V_{ca} \times M_{H_2O}}{R \times T} \quad (10)$$

where $W_{o_2_con}$ stands for the consumed oxygen. The total flow rate at the cathode outlet can be calculated by^[8]

$$W_{ca_out} = C_{ca_out} (P_{ca} - P_{rm}) \quad (11)$$

The pressure ratio of oxygen and nitrogen in the cathode channel can be calculated by using equation (1), then the outlet flow rate of oxygen and nitrogen can be calculated by the combination of equation (10) and equation (11).

The condenser is just the opposite of the humidifier. The flow rate at the outlet can be expressed as

$$W_{con_out} = W_{ca_out} \times \frac{(x_{o_2_out} \times M_{O_2} + (1 - x_{o_2_out} - x_{vap_out}) \times M_{N_2})}{x_{vap_out} \times M_{H_2O} + x_{o_2_out} \times M_{O_2} + (1 - x_{o_2_out} - x_{vap_out}) \times M_{N_2}} \quad (12)$$

Then the oxygen molar fraction of the exhaust gas is exactly the oxygen molar fraction of the gas that comes out

from the condenser. And the molar fraction can be expressed as

$$x_{o_2_rp} = \frac{W_{ca_out}}{W_{con_out}} \times \frac{x_{o_2_out}}{(1 - x_{vap_out})} \times \frac{(x_{o_2_out} \times M_{O_2} + (1 - x_{o_2_out}) \times M_{N_2})}{x_{vap_out} \times M_{H_2O} + x_{o_2_out} \times M_{O_2} + (1 - x_{o_2_out} - x_{vap_out}) \times M_{N_2}} \quad (13)$$

When the molar fraction of oxygen is calculated, the molar mass of return pipe gas can be expressed as

$$M_{rp_air} = \frac{1}{(1 - x_{vap_out})} (x_{o_2_out} \times M_{O_2} + (1 - x_{o_2_out}) \times M_{N_2}) \quad (14)$$

After getting the molar mass of the return pipe gas, the molar mass of the mixed gas that enters the air compressor mentioned before can be expressed as

$$M_{mixair} = \frac{W_{cp}}{\frac{W_{cp} - W_{rp}}{M_{air}} + \frac{W_{rp}}{M_{rp_air}}} \quad (15)$$

C. Stack voltage model

The stack voltage model presented in [8] is used for calculating the voltage in this paper. Generally, the stack voltage V_{cell} is made of four parts: open circuit voltage E_{cell} , activation loss V_{act} , ohmic loss V_{ohm} and concentration loss V_{conc} . The expression can be written as^[8]

$$V_{cell} = E_{cell} - V_{act} - V_{ohm} - V_{conc} \quad (16)$$

The open circuit voltage is slightly affected by temperature and gas pressure. The activation loss is a function of current density. The ohmic loss is determined by current density and temperature. The concentration loss is highly related to the gas pressure and gas component. The detailed calculation is presented in [8].

III. CONTROL STRATEGY

Based on the model described in Part II, a control method showed in figure 5 was conducted to achieve the goal of lowering the voltage.

According to the current demand I , the mass flow rate of the compressor W_{cp} and the pressure at the outlet of the compressor P_{cp} , the total opening area of two throttles can be calculated. The object of this calculation is to maintain the pressure of the air supply system.

When the cell voltage is higher than the reference voltage V_{ref} , the control strategy would be activated. First, a look up table is used for calculating the feedforward recirculation rate. And the recirculation rate will be modified by a PI controller,

whose input is the difference between the real cell voltage and the reference voltage.

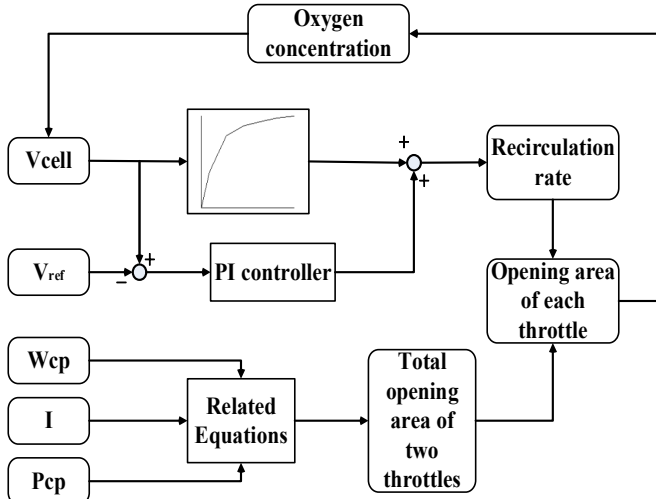


Fig. 5 Control strategy of lowering the voltage

On the other hand, a PI controller based on the difference of reference voltage and the real time voltage V_{cell} is used to guarantee the control accuracy of the strategy.

IV. SIMULATION RESULT

In this part, two conditions are taken into consideration. One is called small load condition and the other is surge condition. The small load condition, which has been described in Part I, is the most common reason of causing high voltage. The surge condition occurs when the fuel cell stack is suffering bleeding. Then the voltage would rise because of the increase of air supply, and sometimes the voltage would be higher than the reference voltage.

A typical result of small load condition is showed in Fig.6. Subfigure1 represents the decrease of air supply, which is equivalent to the decrease of current density. When the current density reduces from $0.3A/cm^2$ to $0.1A/cm^2$, the average cell voltage, which is represented by the green line in subfigure3, will be higher than $0.8V$, which is the reference voltage, without EGR.

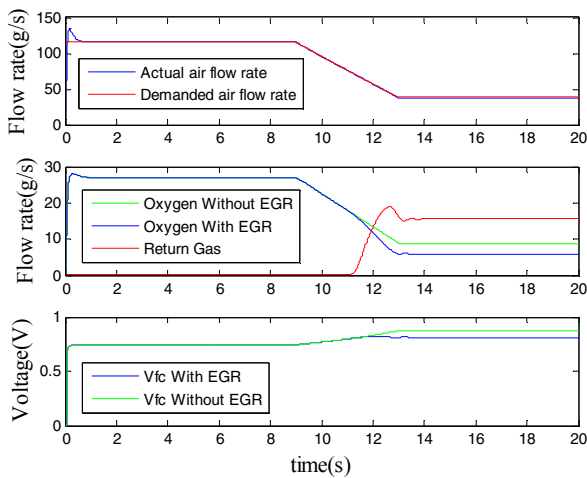


Fig.6 Small load condition

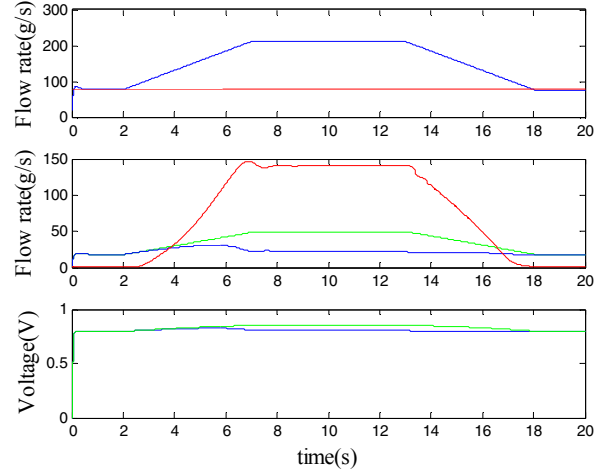


Fig.7 Surge condition

Once when the average cell voltage reaches $0.8V$, the control strategy is activated. The EGR throttle opens and the flow rate of return gas rises, as showed by the red line in subfigure2. At the same time, the flow rate of the oxygen will be lower than that without EGR, thus causing the fallen of the average cell voltage. As the load continued to decrease, the flow rate of return gas keep rising, in order to keep the average cell voltage below $0.8V$. When the load stop changing, the flow rate of return gas will be a constant that makes sure the average cell voltage is below $0.8V$.

Fig.7 shows a typical result of surge condition. The meaning of each line is the same to Fig.6. When surge occurs at a relatively small load, the average cell voltage may exceeds the reasonable range. As showed in subfigure1, when the stack suffers bleeding, the actual air supply increase to a large amount ignoring the fact that the load of the stack keeps a constant. Since the oxygen supply is far more than the demand, the flow rate of return gas needs to be large, as showed by the red line in subfigure2. In subfigure3, the gap between two voltage line is not so significant because when oxygen is supplied to a certain extent, the effect of oxygen stoichiometric to average cell voltage is so small.

In these two conditions described above, it can be clearly seen that the model runs well and the average cell voltage can be controlled below $0.8V$.

V. CONCLUSION

It has been confirmed that the high voltage of PEMFC during small loading is harmful to the durability, and the voltage is highly related to the oxygen stoichiometric. Getting the oxygen stoichiometric under control is the key of lowering the voltage.

Based on the concept described above, an air supply system model with EGR has been built. The simulation results showed that the apply of EGR can greatly contribute to lowering the output voltage, thus improving the durability of PEMFC.

A feasible control strategy is proposed in part III, the strategy can guarantee that the voltage is limited to the reference voltage.

Simulation result of tow typical conditions are presented in part IV. Results showed that the model runs well and the strategy can actually make the average cell voltage lower than 0.8V.

However, there is still more work to do. First, the stack voltage model should be improved. It can only reflex the effect of oxygen pressure but not the effect of oxygen stoichiometric. Then after the improving of the voltage model, the control strategy could be accurately verified using the model. And some experiment should be done to further confirm the model.

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